
IC 131: Applied Chemistry for Engineers

Course Code	IC 131
Course Name	Applied Chemistry for Engineers
L-T-P-C	2-0-2-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Institute Core for All
Approval	recommended by the 38th BoA (Jan 2021)

Course Contents

Part I – Classical mechanics

- Chemical Bonding; MO Theory; LCAO molecular orbitals; structure, bonding and energy levels of diatomic molecules, 3D, 2D, 1D and 0D materials. [7 Lectures]
- Intermolecular Forces; Potential energy surfaces-Rates of reaction; Steady state approximation and its applications; Catalysis. [4 Lectures]
- Spectroscopy- Introduction and Classification; Basic Principles, instrumentation and technological applications of - Ultra Violet - Visible Spectroscopy; Infra-red Spectroscopy; Raman Spectroscopy; and Nuclear Magnetic Resonance Spectroscopy [10 Lectures]
- Electrochemistry and its applications in Fuel Cells; Batteries; and Supercapacitors [7 Lectures]

Experiments illustrating the concepts of:

- Chemistry in Real Life: analysis of food, soil and water quality.
- Synthesis of materials and their characterization using analytical tools.
- Electrochemistry.

Text Books

1. Wiley India Editorial Team, **Engineering Chemistry**, Wiley India Pvt. Ltd., 2011.
2. Shashi Chawla, **Engineering Chemistry**

References

1. Colin Banwell and Elaine McCash, M A Ruderman, **Fundamentals of molecular spectroscopy**, Tata McGraw Hill Education Pvt. Ltd.
- V. S. Bagotsky, **Fundamentals of Electrochemistry**, John Wiley and Sons Inc., 2005.